

Project Design Phase

Problem – Solution Fit Template

Project Name	Rising Water
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Problem – Solution Fit Template:

Problem

Floods are among the most destructive natural disasters, causing significant loss of life, property damage, and economic disruption every year. Existing flood forecasting systems often rely on traditional statistical methods or manual analysis, which may not accurately capture complex weather patterns. These systems frequently produce delayed or inaccurate predictions, leaving disaster management authorities with limited time to issue warnings, plan evacuations, and allocate emergency resources. As a result, communities in flood-prone areas remain highly vulnerable, and emergency response efforts become less effective.

Solution

To address these challenges, we developed a Machine Learning-Based Flood Prediction System that predicts the likelihood of flood events using historical meteorological data. The system analyzes key weather parameters such as annual rainfall, seasonal rainfall, and cloud visibility using multiple classification algorithms, including Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost. After evaluating all models, XGBoost was selected as the best-performing algorithm, achieving 96.55% prediction accuracy.

The trained model is integrated into a Flask web application, providing an intuitive interface where meteorologists and disaster management authorities can enter current weather conditions and instantly receive flood risk predictions. The application supports early warning generation, evacuation planning, disaster preparedness, and efficient resource allocation. Designed for deployment on IBM Cloud, the solution is scalable, accessible from any location, and enables faster, data-driven decision-making to reduce the impact of floods on communities.

Purpose:

- ☐ Solve a critical real-world problem by providing an accurate and timely flood prediction system that helps disaster management authorities make informed decisions.
- ☐ Improve adoption of the solution through an easy-to-use Flask web application that enables meteorologists and government agencies to access flood risk predictions quickly.
- ☐ Enhance disaster response by enabling better resource planning and reducing the loss of lives and property through reliable flood risk predictions.
- ☐ Understand historical weather patterns using machine learning to improve flood prediction accuracy and support data-driven disaster management strategies.
- ☐ Provide early flood warnings that enable authorities to take proactive measures, reducing the impact of floods on communities.
- ☐ Reduce dependence on traditional forecasting methods by leveraging machine learning models to deliver faster and more accurate flood risk predictions.
- ☐ Support timely evacuation planning and efficient emergency resource allocation.

Template:

PROBLEM-SOLUTION FIT CANVAS									
Purpose / Vision: To build an intelligent and accurate flood prediction system using machine learning that provides early warnings, enables timely action, and minimizes the impact of floods on lives and property. /									
1 CUSTOMER SEGMENT(S) Who is your customer? - Disaster Management Authorities - Government Agencies - Meteorologists - Disaster Relief Coordinators - Local Administration in Flood-Prone Areas - Environmental Monitoring Organizations	2 JOBS-TO-BE-DONE / PROBLEMS (J&P) Which jobs-to-be-done (or problems) do you address? - Predict flood occurrence before it happens. - Provide early flood warnings. - Support evacuation planning. - Help authorities allocate emergency resources. - Reduce loss of life and property. - Improve disaster preparedness.	3 TRIGGERS What triggers customers to act? - Heavy rainfall - Continuous rainfall over several days - Monsoon season - Weather department alerts - Low cloud visibility - Rising flood risk indicators - Previous flood incidents	4 EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? Before: - Uncertainty - Anxiety - Fear of flooding - Lack of preparedness - Confusion in decision-making After: - Confidence - Better preparedness - Faster decision-making - Improved public safety - Trust in flood predictions	5 AVAILABLE SOLUTIONS Which solutions are available to the customers? What have they tried in the past? What pros & cons do these solutions have? - Traditional weather forecasting - Manual rainfall analysis - Government warning systems - Satellite monitoring - Hydrological forecasting models Limitations: - Lower prediction accuracy - Delayed warnings - Manual analysis - Slow response time - Limited accessibility	6 CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? - Limited real-time weather information - Budget constraints - Short decision-making time - Dependence on traditional forecasting - Limited technical expertise - Inadequate prediction accuracy	7 BEHAVIOUR What does your customer do to address the problem and get the job done? - Monitor weather reports daily - Collect rainfall and weather data - Coordinate with meteorological departments - Analyze historical flood records - Issue emergency alerts - Deploy rescue teams after receiving warnings	8 CHANNELS OF BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? - Flask Web Application - IBM Cloud Platform - Government Disaster Management Portals - Weather Data Websites - Email Notifications - Online Monitoring Dashboards 8.2 OFFLINE What kind of actions do customers take offline? - Disaster Control Rooms - Emergency Meetings - Field Surveys - Public Warning Announcements - Rescue Team Coordination - Local Administration Offices	9 PROBLEM ROOT CAUSE What is the real reason that this problem exists? Traditional flood forecasting systems rely heavily on manual analysis and conventional prediction methods, which often fail to process complex weather patterns efficiently. This results in delayed and less accurate flood predictions, reducing the time available for authorities to respond effectively.	10 YOUR SOLUTION If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. Our Solution: A Machine Learning-based Flood Prediction System using Decision Tree, Random Forest, KNN, and XGBoost algorithms. The XGBoost model achieved 96.55% accuracy and is integrated into a Flask web application for real-time flood risk prediction. The system enables early warnings, evacuation planning, and efficient disaster resource allocation. Designed for scalable deployment on IBM Cloud.
TR - Triggers SL - Solution CH - Channels EM - Emotions J&P - Jobs & Problems		TR → SL → CH → EM							